

QA and Integration Working Update

Sprint delivery, Thursday 30 July 2026

INTERN	Dawit Worku
WORKSTREAM	Integration and QA lead — Integration Team and Security & QA Team, brief §12
REPOSITORY	async-arch/Async_school_system.
BRANCH AT SUBMISSION	main @ ff88530
TEST RESULT	40 tests executed, 0 failed, 0 errors of 34 test methods
PULL REQUESTS MERGED	#11, #14, #15, #16, #17

1 Role and scope

Section 12 of the brief assigns two workstreams to this role. The Integration Team owns attendance and mark-list teacher permissions, schedule links, relational consistency and regression testing. The Security and QA Team owns access groups, record rules, document privacy, conflict tests, cross-role tests, demo data, the bug log and final integration.

In practice the work was to act as the gate on the protected branch: audit every incoming pull request against the brief, find and fix defects before they reached `main`, and build the security layer that the other workstreams' acceptance criteria depend on. Eighteen defects were logged across the sprint, sixteen of which are fixed.

2 Principal finding

The security layer specified in section 11 did not exist. The file `security/school_security.xml` contained an empty `<odoo/>` element — no access groups and no record rules — and all eighteen access-control rows were bound to the generic `base.group_user` and `base.group_system`.

This mattered because roughly a third of the non-negotiable acceptance criteria in section 13 are statements about what a given role can and cannot see. None of them could be satisfied, and demonstration steps 11 through 15 in section 14 could not be performed at all.

Two further modules named in section 2 as part of the existing foundation were also found to be hollow rather than merely incomplete.

MODULE	STATE FOUND
Mark List	<code>models/school_mark.py</code> was a zero-byte file that was nonetheless imported by the package. Its view file contained an empty <code><odoo/></code> . The model did not exist.
Attendance	The model existed at 48 lines, but its view file was empty and no menu item pointed at it, so the module was unreachable through the interface.

3 Defect log

Severity is recorded as S1 where a defect blocks a stated acceptance criterion, S2 where it breaks a working flow, and S3 for hygiene and cosmetic issues.

#	SEV	DEFECT	FOUND DURING	STATUS
1	S1	No access groups or record rules; section 13 role isolation unsatisfiable	Pre-merge audit	Fixed
2	S1	The <code>school.mark</code> model was absent entirely	Pre-merge audit	Fixed
3	S1	Attendance had no views and no menu entry	Pre-merge audit	Fixed
4	S1	Responsibility Assignment, a mandatory module, did not exist. Non-teaching staff could hold no responsibility, so responsibility-targeted announcements could never reach a Registrar or a Librarian	Brief re-audit	Fixed
5	S1	<code>school.teacher.name</code> was an independent character field, so renaming the staff master record did not propagate — a direct failure of the section 13 criterion on master-record renames	Brief re-audit	Fixed
6	S1	Pull request #11 introduced three models with no access-control rows at all; every non-administrator would have hit an access error	PR #11 review	Fixed before merge
7	S1	Staff and teacher required documents had no fields on either model	Brief re-audit	Fixed
8	S2	The <code>staff_id</code> NOT NULL migration fails on any database that already holds teacher rows; Odoo logs the failure and silently leaves the column nullable	Post-merge upgrade, PR #14	Open
9	S2	Demo data sat below the section 13 minimums, at three staff and two teachers against the required five and three	Acceptance-criteria check	Fixed
10	S2	No student records existed in demo data, so attendance and marks could not be exercised at all and demonstration steps 13 and 15 dead-ended	Browser test preparation	Fixed
11	S2	The administrator overview rendered every tile as zero against a database holding five active staff, caused by a blocked transient record and a compute with no field dependencies	Browser verification	Fixed
12	S2	The test suite passed only against an empty database; nine of nine tests errored once real data existed because fixture values collided with seeded records	Post-seed test run	Fixed
13	S2	The administrator password is not <code>admin</code> on databases created from the command line, so the documented login was wrong	Browser test pass	Fixed
14	S3	Unsaved class schedules displayed <code>False - False (00:00)</code> in the breadcrumb	Browser verification	Fixed
15	S3	Merge conflict in the manifest between pull request #11 and the main branch	PR #11 merge	Resolved
16	S3	The manifest declared no licence key, producing an Odoo warning on every load	Install log review	Fixed
17	S3	Two fields on <code>school.staff</code> shared the label "Responsibilities"	Upgrade log review	Fixed
18	S3	Database credentials committed in a tracked configuration file, with the database manager left unprotected	Security review	Open

4 Integration decisions and their reasoning

Four decisions shaped the final implementation. Each is recorded here with its reasoning, because section 15 requires being able to explain the configuration shown.

Required fields enforced by constraint rather than by `required=True`

Section 4 requires a name, phone, department, job title, responsibility and employment status on every staff record. Declaring these `required=True` adds a NOT NULL constraint to a column that already holds rows, which is precisely the migration that failed in defect 8. A Python constraint gating the move out of Draft gives the same guarantee without a migration that can fail on an existing database.

A control status added alongside employment status, not replacing it

Section 4 describes a control status of Draft, Active, Suspended, Inactive or Archived, and separately an employment status. Overwriting the existing selection would have silently reinterpreted data owned by the Staff Registration team. Two fields were kept, with two distinct meanings.

Documents and responsibilities added through inheritance in separate files

The Staff Registration and Teacher Registration teams' own model and view files were left untouched, so their in-flight pull requests would not conflict with the security work.

Record-rule domains reduced to plain attribute lookups

Odoo evaluates record-rule domains through `safe_eval`, which permits attribute access but not method calls, and which has no `datetime` in scope. Both limits were discovered during the first install attempt, which failed with `name 'datetime' is not defined`. Each user's scope is therefore precomputed onto fields on `res.users` so that every rule domain is a simple attribute lookup.

One earlier assessment was withdrawn. Branch and campus targeting was initially reported as not implementable, because nothing in the module modelled a campus. On re-audit a small `school.campus` model made it straightforward, and all eight audience types specified in section 8 now function.

5 Merge gatekeeping

PR	SUBJECT	AUTHOR	ACTION TAKEN
#11	Teacher, subject and assignment models	team	Manifest conflict resolved and the six missing access-control rows added before the merge was allowed
#14	Teacher to staff link, workload validation	mekhlua	Reviewed and merged with continuous integration green. The post-merge upgrade surfaced defect 8, which was reported rather than concealed
#15	Program and class scheduling	self	Merged after integration and browser verification
#16	Security, marks, announcements, dashboards	self	Merged after a full test run and three-role browser verification
#17	Demo students, attendance and marks	self	Merged
#9	Mark list	Jennah198	Held, not merged. The branch is in conflict and now overlaps the <code>school.mark</code> model delivered in #16. Flagged for rebasing rather than conflict resolution

6 Test evidence

```
docker compose exec odoo odoo -c /etc/odoo/odoo.conf -d <db> -u school_management \
  --test-enable --test-tags /school_management --no-http --stop-after-init

school_management: 40 tests 0 failed 0 error(s) of 34 tests
```

The suite comprises 461 lines across three files and 34 test methods. It was verified clean against three separate conditions: a fresh installation with demo data, a fresh installation without demo data, and an upgrade applied to an existing populated database.

FILE	TESTS	COVERAGE
<code>test_class_schedule.py</code>	9	Section 7.3 conflict validation — teacher, class and room double-booking; back-to-back slots permitted; the same weekday in another term permitted; a cancelled slot releasing its room; an unassigned teacher rejected; a weekday or date required; an inactive teacher blocked from publication; a reason required when rescheduling
<code>test_security.py</code>	12	Sections 9 and 13 isolation — a teacher sees attendance, marks and students only for assigned classes and subjects; announcements aimed at another class or subject are invisible; draft and expired announcements are invisible; staff documents raise an access error below Registrar; a plain staff user sees no other staff records
<code>test_responsibility.py</code>	13	Sections 4, 5 and 6 — a staff rename propagating to the teacher and the assignment label; the Draft gate; suspended staff refused new assignments; deactivation disabling the linked login; one primary responsibility per person; no self-reporting; one homeroom teacher per class and term; responsibility and campus announcement targeting

7 Demonstration evidence

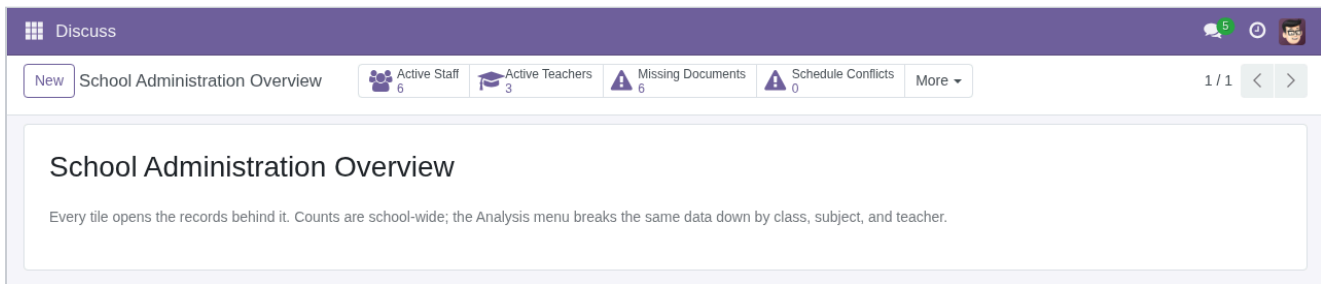


Figure 1 — Administrator overview, section 10. Counts are read live from the database and each tile opens the records behind it. Schedule conflicts read zero because the create and write constraints reject clashes at source. This screen previously rendered every tile as zero; that was defect 11. The sixth staff record was added during manual testing after the demo data shipped with five.

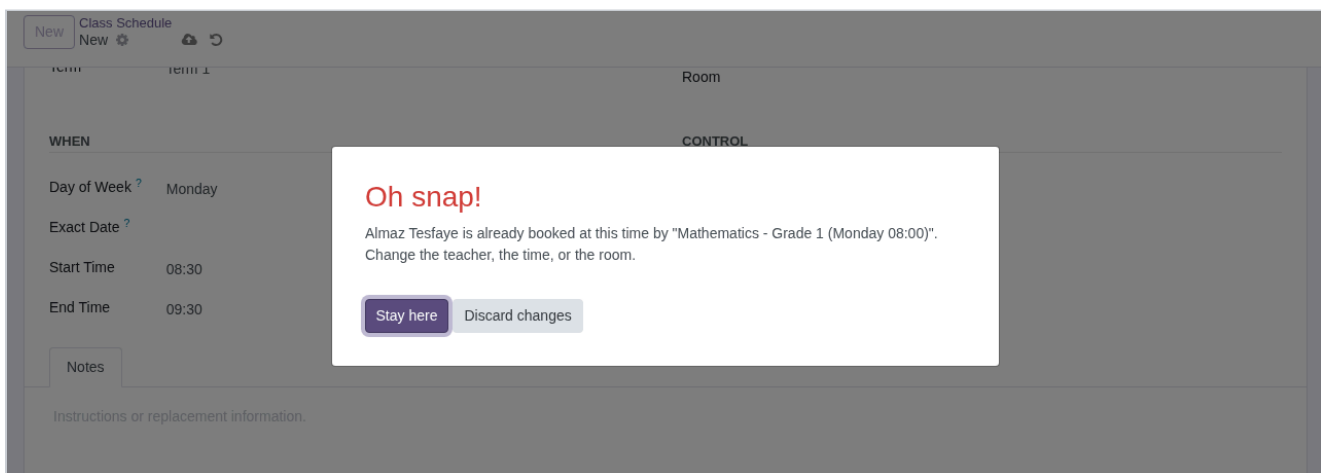


Figure 2 — Conflict validation, section 7.3. An attempt to schedule Almaz Tesfaye for Grade 2 A on Monday from 08:30 to 09:30 is rejected because she already teaches Grade 1 A from 08:00 to 09:00. The message names the clashing record. This is a backend constraint raised on create, not a user-interface warning, which section 7.3 states explicitly is not sufficient on its own.

Role isolation, section 13

The three figures below show the same menu, My Announcements, opened by three different users against the same database. Nothing about the view changes; only the record rules differ.

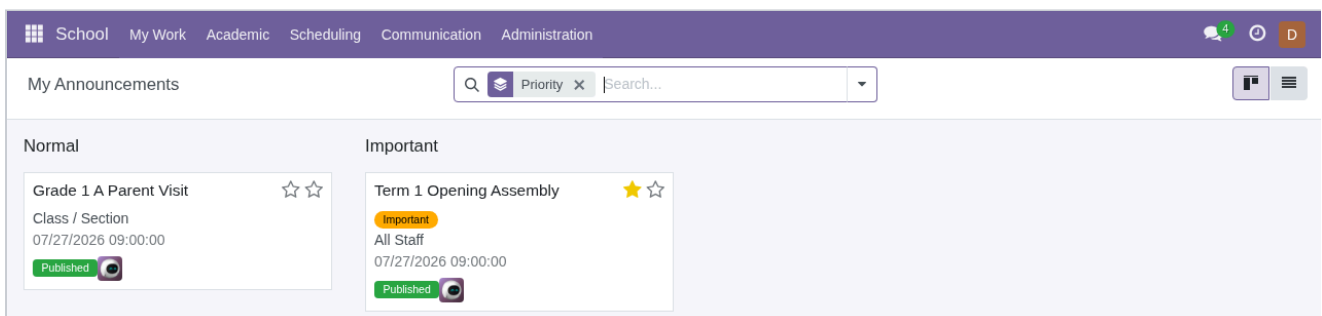


Figure 3 — Mathematics teacher. Sees the all-staff assembly notice and the Grade 1 A parent visit, because she teaches that class. She does not see the laboratory closure, which is targeted at General Science, nor the registration notice targeted at the Registrar responsibility.

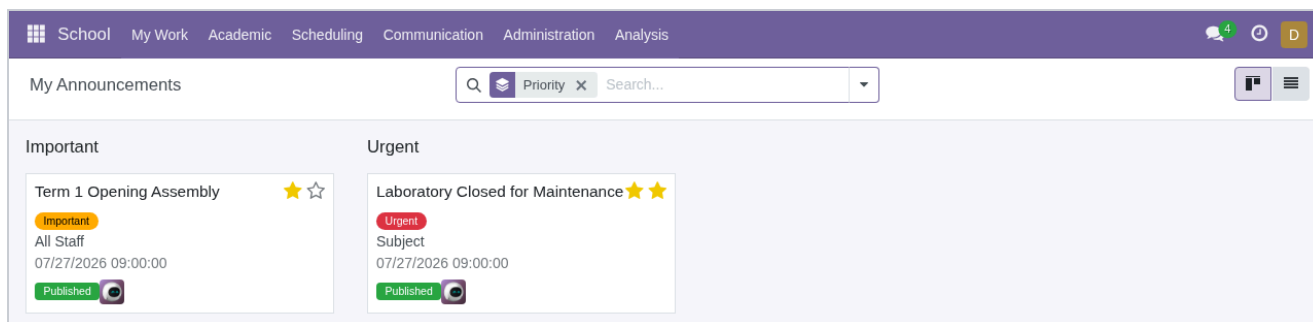


Figure 4 — Science teacher, the exact complement. Sees the all-staff notice and the urgent laboratory closure for General Science, and does not see the Grade 1 A parent visit. The urgent item carries a visible priority indicator, as section 8 requires. This user also holds Department Head, which is why the Analysis menu appears here and not in Figure 3.

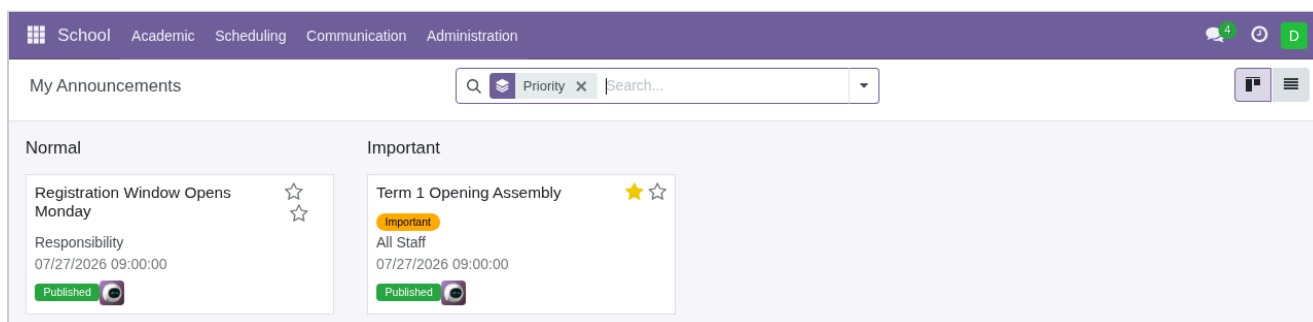


Figure 5 — Registrar, a non-teaching staff member. Sees the registration notice targeted at the Registrar responsibility, and neither class-targeted nor subject-targeted item. This case was impossible before the Responsibility Assignment module was built, because responsibility existed only on teaching assignments and a Registrar could therefore hold none. It is the clearest evidence that defect 4 is closed.

8 Open defects and risks carried forward

The staff link constraint is not enforced at database level on upgraded systems

Making `staff_id` required on a table that already held teacher rows fails. Odoo logs the failure and continues, leaving the column nullable. Fresh installations receive the constraint; upgraded ones do not, so on any existing database the link is enforced only by the object layer. The fix is a pre-initialisation migration that backfills existing rows before the constraint is applied. This belongs to the Teacher and Assignment workstream.

Database credentials are committed to the repository

The tracked configuration file carries both the database manager password and the database user password, and the ignore file does not cover it. With the manager unprotected, anyone able to reach the service port can drop or duplicate any database. Section 18 forbids committing database credentials. The remedy is to move both values into the ignored environment file and ship an example configuration in place of the real one. It was not done during the sprint because it changes how the stack starts for every intern.

Section, academic year and term are not modelled as records

Section 6 asks for relational fields to these. Section remains a character field on the class record, academic year a character field, and term a two-value selection. Everything keys off the class record, so the links stay internally consistent, but there is no master table for either concept. This is a schema change touching every workstream and was deliberately not attempted mid-sprint.

Two lesser items

Dated and recurring schedule slots are not cross-checked, so a one-off makeup class landing on a recurring weekday is not flagged; this is marked in the code with its upgrade path. There is also no approval workflow on attendance or marks, which remain direct entry.

A breaking change was shipped deliberately

Access-control rows no longer grant anything to the generic internal user group. Any existing user without a School Management role will see an empty application after upgrading. The Odoo administrator account is added to the School Administrator role automatically; every other account needs a role assigned. This will affect each intern's local database as soon as they pull the main branch, and the team should be told before the demonstration.

9 Deliverables produced

- Role and permission matrix covering seven roles and their implication ladder
- Status transition documentation for schedules, programs and announcements
- A regression checklist of 24 rows spanning registration, attendance, marks, staff, teacher, scheduling and role isolation
- Conflict validation and access-control test evidence, as set out in section 6 above
- A known-issues and incomplete-work list matching section 8 above
- Installation and upgrade commands, including the command-line administrator password caveat found as defect 13
- Demonstration data covering five staff, three teachers, four students, two classes, three subjects, five class periods, two programs, five announcements, eight attendance records, eight marks, two campuses, three rooms and six role logins

10 Recommended next actions

1. Raise an issue for the staff link backfill migration before anyone upgrades another populated database.
2. Notify the team about the access-control breaking change ahead of the demonstration.
3. Tell Jennah198 that pull request #9 needs rebasing onto the merged mark model rather than a conflict resolution.
4. Decide as a team whether academic year and term become master tables. This is the last structural gap against section 6.
5. Move the committed credentials out of the tracked configuration once the demonstration is complete.